

Goal

Implement an algorithm that calculates the value of a portfolio for a given investor on a given reference date.

Description

Each investor may hold multiple types of investments such as stocks, real estate, and funds. Funds themselves may contain various investments, including other funds. The portfolio value is the sum of the values of all individual investments on a specific reference date.

- For a stock, the value equals the number of invested shares multiplied by the stock price on the reference date.
- For real estate, the value equals the sum of estate value and building value (transactions contain absolute values).
- For a fund, the value equals the invested percentage share multiplied by the total value of the fund's contained investments.

Input

Three CSV files are provided as input. Data may not be ordered logically and should be processed accordingly.

- Investments: Contains assignments of investors to investments and the investments of funds.
- Transactions: Contains transactions of investments (shares, estate value, building value, percentage).
- Quotes: Contains historical stock prices (prices may not be available for every day).

Task

Write a console application in C# (or Java) that reads investor ID and reference date from console input and calculates the value of the investor's portfolio on that date.

Input format example: YYYY-MM-DD;InvestorId